

Assignment 3

Date: 19th December, 2025

Instructions: Perform a complete dry run of the program on paper, take clear pictures of the working steps, and compile them into a PDF file for assignment submission.

Q#01

Zameen.pk is introducing a “**Smart Movers**” feature that helps customers **select the best items to load into their moving truck** (limited capacity) while shifting to a new home.

You are shifting to a new house in Lahore.

Zameen.pk provides you with a **moving truck** that can carry a **maximum weight of 200 kg**.

You have several household items, each with a:

- **Weight**
- **Value (importance to take first)**
(Value could be usefulness, replacement cost, emotional value, etc.)

Your goal is to **maximize the total value** of items loaded into the truck **without exceeding 200 kg**.

Item	Weight (kg)	Value (points)	Description
Sofa Set	80	120	Premium 3-seater, expensive to replace
LED TV	20	90	High value, fragile, compact
Refrigerator	60	150	Essential appliance
Double Bed	70	100	Important, but heavy

Laptop	5	70	Extremely important, high value
Dinner Set	15	40	Useful but replaceable
Clothes Box	10	30	Medium importance

Just like Zameen.pk recommends the **best properties** based on user needs, you need to select the **best combination of items** that:

- ✓ Maximizes **value**
- ✓ Fits within **200 kg limit**

This is a **0/1 Knapsack Problem**:

- You either **take an item completely or leave it**
- No partial items allowed (you can't take half a sofa)

Q#02

Emergency 1122 receives emergency calls.

Each call is assigned an **integer priority**:

- **Higher integer = more critical emergency**
- The system always dispatches the ambulance to the **highest priority integer** first
- This is exactly how a **Max Heap (Priority Queue)** works

Incoming Calls (Represented as Integer Priorities)

Let the calls arrive with the following priority levels:

25, 40, 10, 60, 35

Where:

- **60** = cardiac arrest (highest urgency)
- **40** = severe accident
- **35** = fire incident

- **25** = injury
- **10** = minor issue

Write an ambulance dispatch algorithm which will dispatch the ambulance on the basis of maximum priority of integers in $O(\log n)$